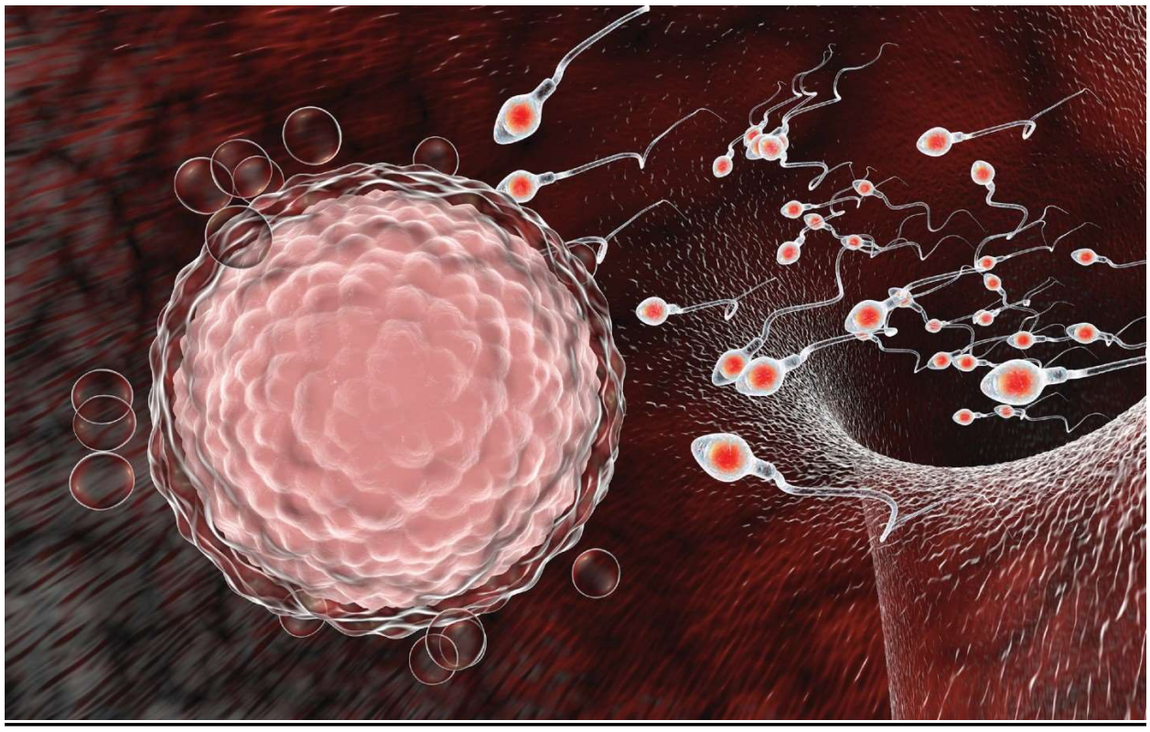
ZOO501- DEVELOPMENTAL BIOLOGY

Final-Term File 2023 By

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Composition of hyaline membrane:

The hyaline layer is actually made up of two layers, an outer lamina made primarily of hyalin protein and an inner lamina composed of fibropellin proteins.

Which type of egg present in birds or chicks.

Telolecithal egg is present in birds. It is a type of egg that has large amount of yolk that is unevenly distributed in the cytoplasm.

In which portion the hemisphere present.

The portion of egg that has large amount of yolk is known as vegetal hemisphere and it is present toward posterior side of the egg.

Write the three name of vesicle formed in brain.

1= Prosencephalon (forebrain), 2= Mesencephalon (midbrain), 3= Rhombencephalon (hindbrain).

What are the osteoclasts:

Osteoclasts are the cells that degrade bone to initiate normal bone remodeling and mediate bone loss in pathologic conditions by increasing their resorptive activity. They are derived from precursors in the myeloid/ monocyte lineage that circulate in the blood after their formation in the bone marrow.

Differences between identical and fraternal twins.

Identical twins	Fraternal twins
Identical twins are monozygotic twins.	Fraternal twins are dizygotic twins.
They are formed from a single zygote or fertilized egg.	They are formed from two eggs are fertilized by two different sperms.
The zygote splits into two parts and each part develops into an individual embryo.	Two eggs are fertilized by two different sperms.
Their genetic origin is the same and therefore they have the same DNA.	They share 50% of their genetic traits, just like any other sibling born at different times.
They are of the same sex.	The sexes may be the same or different.
They share the same blood type.	The blood types may or may not be the same.

What are the three axis of foundation of the body:

Three axes are the foundations of the body: the anterior-posterior axis (head to tail or mouth to anus), the dorsal-ventral axis (back to belly), and the right-left axis (between the two lateral Sides of the body).

Write a note on osteoclast in detail:

Osteoclasts are the cells that degrade bone to initiate normal bone remodeling and mediate bone loss in pathologic conditions by increasing their resorptive activity. They are derived from precursors in the myeloid/ monocyte lineage that circulate in the blood after their formation in the bone marrow. The blood vessels also import the blood-forming cells that will reside in the marrow for the duration of the organism's life. The number and activity of osteoclasts must be tightly regulated. If there are too many active osteoclasts, too much bone will be dissolved, and osteoporosis will result. Conversely, if not enough osteoblasts are produced, the bones are not hollowed out for the marrow, and osteopetrosis results.

Differentiation of the neural tube:

The brain develops from this early tube structure and gives rise to specific regions of the adult brain. As the neural tube grows and differentiates, it enlarges into three vesicles that correspond to the forebrain, midbrain, and hindbrain regions of the adult brain. Later in development, two of these three vesicles differentiate further, resulting in five vesicles. Those five vesicles can be aligned with the four major regions of the adult brain. The cerebrum is formed directly from the telencephalon. The diencephalon is the only region that keeps its embryonic name. The mesencephalon, metencephalon, and myelencephalon become the brain stem. The cerebellum also develops from the metencephalon and is a separate region of the adult brain.

Define neural plate:

The neural plate is a key developmental structure that serves as the basis for the nervous system. Opposite the primitive streak in the embryo, ectodermal tissue thickens and flattens to become the neural plate. The region anterior to the primitive knot can be generally referred to as the neural plate.

What is meroblastic Cleavage.

Meroblastic cleavage involves an early separation between cells at the animal pole and yolk at the vegetal pole of the egg, and cell furrows encompass only the animal-most region of the embryo, leaving the yolk intact. For example: cleavages of birds, insects, fish, and reptiles.

Define sub-germinal cavity and its function:

Between the blastoderm and the yolk is a space called the subgerminal cavity. This space is created when the blastoderm cells absorb fluid from the albumin (“egg white”) and secrete it between themselves and the yolk. It gives the central part of the blastodisc a semitransparent appearance and this area is called the area pellucida.

Define diploblastic animals and their examples:

Diploblastic animals are those who have ectoderm and endoderm, but no true mesoderm. These include the cnidarians (jellyfish and hydras).

Events of primary neurulation:

In primary neurulation, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surface to form a hollow tube.

The head and trunk regions both undergo variants of primary neurulation, and this process can be divided into four distinct but spatially and temporally **overlapping stages**: (1) formation of the neural plate; (2) shaping of the neural plate; (3) bending of the neural plate to form the neural groove; and (4) closure of the neural groove to form the neural tube.

Function of placenta:

Placenta is a disc like structure that forms a connection between the embryo and the uterine wall. The placenta acts to provide oxygen and nutrients to the fetus, whilst removing carbon dioxide and other wasteproducts. It metabolises a number of substances and can release metabolic products into maternal and/or fetal circulations.

Blastula formation in Tunicate:

These animals are characterized by bilateral holoblastic cleavage, a pattern found primarily in tunicates. Each successive division orients itself to this plane of symmetry, and the half-embryo formed on one side of the first cleavage plane is the mirror image of the half-embryo on the other side. The second cleavage is meridional, like the first, but unlike the first division, it does not pass through the centre of the egg. Rather, it creates two large anterior cells (the A and D blastomeres) and two smaller posterior cells (blastomeres B and C). Each side now has a large and a small blastomere. During the next three divisions, differences in cell size and shape highlight the bilateral symmetry of these embryos. At the 32-cell stage, a small blastocoel is formed, and gastrulation begins.

Classes of twins:

Human twins are classified into two major groups: monozygotic (one egg, or identical) twins and dizygotic (two egg, or fraternal) twins.

Factors in myogenesis:

The Myogenic Regulatory Factors (MRFs) Myf5, MyoD, myogenin and MRF4 are members of the basic helix-loop-helix family of transcription factors that control the determination and differentiation of skeletal muscle cells during embryogenesis and postnatal myogenesis.

Primary vesicles of the brain:

Initially there are three primary brain vesicles: prosencephalon, mesencephalon, and rhombencephalon.

Why ascidian are called tunicates:

Ascidiacea, commonly known as the ascidians, tunicates (in part), and sea squirts (in part), is a polyphyletic class in the subphylum Tunicata of sac-like marine invertebrate filter feeders. Ascidians are characterized by a tough outer "tunic" made of a polysaccharide.

They are known as Tunicate because the body of an adult is enclosed within a tunic made of a polysaccharide.

Tunicate gastrulation:

Tunicate gastrulation is characterized by the invagination of the endoderm, the involution of the mesoderm, and the epiboly of the ectoderm. About 4–5 hours after fertilization, the vegetal (endoderm) cells assume a wedge shape, expanding their apical margins and contracting near their vegetal margins. Pairs of blastomeres appear to lead this invagination into the centre of the embryo. This invagination forms a blastopore whose lips will become the mesodermal cells.

How neural grove formed in primary neurulation:

Shortly after the neural plate has formed, its edges thicken and move upward to form the neural folds, while a U-shaped neural groove appears in the center of the plate, dividing the future right and left sides of the embryo.

What is the Henson node:

At the anterior end of the primitive streak is a regional thickening of cells called the primitive knot or Hensen's node.

Difference between mantile zone and ventricular zone:

As the cells adjacent to the lumen continue to divide, the migrating cells form a second layer around the original neural tube. This layer becomes progressively thicker as more cells are added to it from the germinal neuroepithelium. This new layer is called the mantle (or intermediate) zone, and the germinal epithelium is now called the ventricular zone (and, later, the ependyma).

which factor is responsible for cartilage specification:

Pax1 and Scleraxis are the transcription factors are thought to activate cartilage-specific genes.

What is neural crest and its function:

A group of ectodermal cells that gives rise to the spinal ganglia and various structures of the autonomic nervous system.The neural crest cells originate at the dorsal most region of the neural tube.The neural crest can be divided into four main functional (but overlapping) domains. The cranial (cephalic) neural crest, whose cells migrate dorsolaterally to produce the craniofacial mesenchyme that differentiates into the cartilage, bone, cranial neurons, glia,and connective tissues of the face.

Type of gastrulation in snails:

The snail stereoblastula is relatively small. Gastrulation in Snails is accomplished primarily by epiboly, wherein the micromeres at the animal cap multiply and "overgrow" the vegetal macromeres. Eventually, the micromeres will cover the entire embryo, leaving a small slit at the vegetal pole. In the typical cleavage of molluscs, cells divide to produce most of the larval structures, including a gland capable of producing a large shell.

What is the umbilical cord:

The umbilical cord connects the baby to the mother's placenta. During fetal development in the womb, the umbilical cord is the lifeline to the baby supplying nutrients. After birth, the cord is clamped and cut. Eventually between 1 to 3 weeks the cord will become dry and will naturally fall off.

How myotubes forms:

During muscle differentiation, myogenic cells fuse to form multi-nucleated myotubes and myofibers, which extend in parallel through the length of the muscle. Contractile myotubes can also form during in vitro culturing of muscle progenitor cells.

Transcription in regulating:

- Differential gene transcription: It regulates that which of the nuclear genes are transcribed into RNA.
- Selective nuclear RNA processing: It regulating which of the transcribed RNAs (or which parts of such a nuclear RNA) enter into the cytoplasm to become messenger RNAs.
- Selective messenger RNA translation: It regulates that which of the mRNAs in the cytoplasm become translated into proteins.
- Differential protein modification: It regulates that which proteins are allowed to remain or function in the cell.

Endochondral ossification:

The mesenchymal cells differentiate into cartilage, and this cartilage is later replaced by bone. The process by which a cartilage intermediate is formed and replaced by bone cells is called endochondral ossification. It begins when mesoderm-derived mesenchymal cells differentiate into chondrocytes. Chondrocytes proliferate rapidly and secrete an extracellular matrix to form the cartilage model for bone.

Name the germ line layers in animals:

The three germ layers are the endoderm, the ectoderm, and the mesoderm. Cells in each germ layer differentiate into tissues and embryonic organs.

Fate of ectoderm:

The developmental fates of the ectoderm, for instance, can be epidermis, central nervous system, sensory organs, and neural crest.

steps of muscles cell fusion:

Muscle cell fusion begins when the myoblasts leave the cell cycle. As long as particular growth factors (particularly fibroblast growth factors) are present, the myoblasts will proliferate without differentiating.

Steps:

Muscle cell fusion is a multistep process involving cell migration, adhesion, membrane remodeling and actin-nucleation pathways to generate multinucleated myotubes. However, molecular brakes restraining cell–cell fusion events have remained elusive.

What is the differences between primary and secondary neurulation:

In primary neurulation, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surface to form a hollow tube.

During primary neurulation, the original ectoderm is divided into three sets of cells. 1= The internally positioned neural tube, which will form the brain and spinal cord. 2= The externally positioned epidermis of the skin. 3= The neural crest cells

In secondary neurulation, the neural tube arises from a solid cord of cells that sinks into the embryo and subsequently hollows out to form a hollow tube.

Secondary neurulation involves the making of a medullary cord and its subsequent hollowing into a neural tube. In frogs and chicks, secondary neurulation is usually seen in the neural tube of the lumbar (abdominal) and tail vertebrae.

Gray and white matter:

the mantle zone, containing the neuronal cell bodies, is often referred to as the Gray matter; the axonal, marginal layer is often called the White matter. In the spinal cord and medulla, this basic threezone pattern of ependymal, mantle, and marginal layers is retained throughout development. The gray matter (mantle) gradually becomes a butterfly-shaped structure surrounded by white matter; and both become encased in connective tissue.

Difference between Cytokinesis and Karyokinesis?

Karyokinesis is the process of division of the nucleus of a cell during the phase of division (mitosis or meiosis).

Cytokinesis is the process of division of cytoplasm of the cell. Cytokinesis divides cytoplasm and organelles including nuclei into its daughter cells.

Define Syncytial specification:

Syncytial specification. Many insects also use a third means, known as syncitial specification, to commit cells to their fates. Here, interactions occur not between cells, but between parts of one cell. In early embryos of these insects, cell division is not complete. Rather, the nuclei divide within the egg cytoplasm.

Define hypoblast:

Hypoblast is the term used to describe the lower layer of cells of the early bilaminar disc, most commonly in avian embryos: this layer is also termed anterior, or distal, visceral endoderm in mouse embryos.

Define sensor:

Humans have 5 main senses: vision, hearing, smell, touch and taste. Our sensors include the eyes, ears, nose, skin and tongue. Additional sensors include temperature sensors, body position sensors, balance sensors and blood acidity sensors.

Amnion sac:

A thin-walled sac that surrounds the fetus during pregnancy. The sac is filled with liquid made by the fetus (amniotic fluid) and the membrane that covers the fetal side of the placenta (amnion).

Function of retinoic acid:

In the retina, retinoic acid acts as a light-signaling neuromodulator and regulates gap junctionmediated coupling of retinal neurons. It has also been implicated in the control of sleep and circadian clock.

Acrosome reaction:

The acrosome reaction that occurs after sperm capacitation, is an exocytotic event induced by a Ca++ influx. It plays an essential role during fertilization, by making spermatozoa able of penetrating the zona and capable of fusing with the egg plasma membrane.

Difference between dextral coiling and sinistral coiling:

Some snails have their coils opening to the right of their shells (dextral coiling), whereas other snails have their coils opening to the left (sinistral coiling).

How secondary hypoblast form in avians?

Gastrulation of the Avian Embryo Shortly thereafter, a sheet of cells from the posterior margin of the blastoderm (distinguished from the other regions of the margin by Koller's sickle, a local thickening) migrates anteriorly to join the polyinvagination islands, thereby forming the secondary hypoblast.

What is embryonic shield:

The embryonic shield is critical in establishing the dorsal-ventral axis in fishes. It can convert lateral and ventral mesoderm (blood and connective tissue precursors) into dorsal mesoderm (notochord and somites), and it can cause the ectoderm to become neural rather than epidermal.

Define neural tube:

The neural tube forms the early brain and spine. These types of birth defects develop very early during pregnancy, often before a woman knows she is pregnant. The two most common NTDs are spina bifida (a spinal cord defect) and anencephaly (a brain defect).

MHP:

In birds and mammals, the cells at the midline of the neural plate are called the medial hinge point (MHP) cells. They are derived from the portion of the neural plate just anterior to Hensen's node and from the anterior midline of Hensen's node.

Causes: Different neural tube defects are caused when various parts of the neural tube fail to close.

Explain the spina bifida:

Spina bifida is when a baby's spine and spinal cord does not develop properly in the womb, causing a gap in the spine. Spina bifida is a type of neural tube defect. The neural tube is the structure that eventually develops into the baby's brain and spinal cord.

Cleavage pattern in sea urchin, nematode and tunicate?

In all four invertebrates described here, cleavage is holoblastic. In the sea urchin,cleavage is radial; in the snail, spiral; in the tunicate, bilateral; and in the nematode, rotational. In some, such as the sea urchin and tunicate, the axes are established at fertilization through determinants in the egg cytoplasm.

Teratology and Evolutionary embryology:

The third anatomical approach to developmental biology is **teratology**, the study of birth defects. These anatomical abnormalities may be caused by mutant genes or by substances in the environment that interfere with development.

Evolutionary Embryology: The study of how changes in development may cause evolutionary changes and of how an organism's ancestry may constrain the types of changes that are possible.